

BDD Scenarios — Schatzsuche Scratchcard

Note: The below test cases are the translated into BDD format which later can be used as. feature file to run automated tests.

Feature: Prize Plan & Statistical Validation

Scenario: Winning class pays the fixed amount it's supposed to

Given a ticket export from a printed batch
When I compare a ticket's winning class to its paid amount
Then the paid amount matches the prize table exactly for that class

Scenario: Class counts match the declared distribution

Given the full 1,830,000-ticket export
When I group all tickets by winning class
Then each class count matches the declared prize plan exactly

Scenario: Ticket count matches the prize plan statement

Given the full ticket export
When I count every row
Then the total is exactly 1,830,000, with the 183,000 free tickets counted as part of that total, not extra

Scenario: Overall odds come out to 1:7.12

Given the full ticket export
When I divide the total ticket count by the number of winning tickets
Then the result rounds to 7.12

Scenario: Top prize odds come out to 1:915,000

Given the full ticket export
When I divide the total ticket count by the number of WC10 tickets
Then the result is exactly 915,000

Scenario: Distribution holds up at different sample sizes

Given random samples of 50,000 and 500,000 tickets
When I compare each sample's per-class counts to the declared plan
Then the gap stays within normal random variation at both sizes

Feature: Rule 1 — Top Prize Field

Scenario: €15,000 field appears on tickets with 1–5 wins

Given a ticket with between 1 and 5 individual wins
When I check for the top-prize field
Then it should be present

Scenario: €15,000 field is missing on a 6-win ticket

Given a ticket with exactly 6 individual wins
When I check for the top-prize field
Then it should be absent

Feature: Free Tickets (Rule 2, Note 2)

Scenario: FREILOS only shows up on a real free-ticket win

Given a ticket that didn't actually win a free ticket
When I check for the FREILOS symbol
Then it should be absent

Scenario: Free-ticket count and spread across classes

Given the full ticket export
When I count and group free-ticket wins by class
Then the total is 183,000, spread across more than one class

Feature: Decorative vs. Prize-Triggering Amounts (Rules 3 & 4)

Scenario: Decorative symbols are roughly evenly spread

Given the full ticket export
When I compare symbol frequency on winning vs. non-winning tickets
Then no symbol should be unusually skewed toward one or the other

Scenario: A decorative amount never triggers a payout

Given a non-winning ticket showing a decorative euro amount
When I try to settle the ticket normally
Then no payout should occur from that field

Scenario: Decorative and real amount fields can't be confused

Given a decorative €20 on a non-winning ticket and a real €20 payout elsewhere
When I check the underlying data behind both
Then they should be structurally distinct, not just visually similar

Feature: Win Count Boundaries (Rule 5)

Scenario: 6 wins is valid

Given a ticket with exactly 6 individual wins
When it's settled
Then it should be accepted, with no top-prize field, and any repeated amount justified by a real extra win

Scenario: No ticket exceeds 6 wins

Given the full ticket export
When I filter for more than 6 individual wins
Then there should be zero results

Feature: Ticket Book Sequencing (Rules 6, 7, 8)

Scenario: No two consecutive tickets are identical

Given a printed ticket book
When I compare each ticket to the next one
Then no two neighbors should share the same layout

Scenario: A prize amount only repeats when justified

Given a ticket with two real wins of the same amount
When I count that amount on the ticket
Then it should appear exactly twice, matching the two wins

Scenario: Exactly 9 consecutive non-winners is fine

Given a run of exactly 9 non-winning tickets in a row
When I check the ticket right after
Then it should be a winner or a free ticket

Scenario: No run of 10+ non-winners ever happens

Given the full ticket export
When I scan every book for non-winning runs
Then none should reach 10 or more

Feature: Symbol Combinations (the core risk)

Scenario Outline: No ticket falls into an invalid symbol/amount combination

Given a ticket's winning status, its WINNER/DOPPLER/FREILOS symbols, and its amount
When I check that combination against what's actually valid
Then it should never land on one of these invalid cases:

- Non-winning ticket showing WINNER, DOPPLER, or FREILOS
- A winning ticket with WINNER but no amount and no free-ticket win (a win paying nothing)
- A winning ticket with an amount but no WINNER symbol at all
- A DOPPLER win where the doubled amount doesn't match a real prize class
- FREILOS combined with DOPPLER (a ticket prize can't be doubled)
- FREILOS present without WINNER
- FREILOS present on a ticket that didn't actually win a free ticket

Feature: Winning Ticket Symbol Logic (Rules 9, 10, 11)

Scenario: WINNER marks every non-DOPPLER win

Given an ordinary win and a free-ticket win, neither via DOPPLER
When I check both for WINNER
Then WINNER should be present on both, with FREILOS only on the free-ticket one

Scenario: DOPPLER always shows exactly half the final prize

Given a ticket won via DOPPLER
When I check the amount shown before reveal
Then it should be exactly half the final prize — €25 shown for a €50 win, €7,500 shown for the €15,000 top prize, no exception

Scenario: FREILOS never appears without WINNER

Given a ticket showing FREILOS
When I check for WINNER on the same ticket
Then it should also be present

Feature: Non-Winning Ticket Integrity (Rules 12–15)

Scenario: No duplicate symbols on a non-winning ticket

Given a non-winning ticket
When I list its symbols
Then all of them should be distinct

Scenario: WINNER, DOPPLER, and FREILOS never appear on a non-winning ticket

Given a non-winning ticket
When I check for any of these three symbols
Then none of them should be present

Feature: Client / Backend Consistency

Scenario: Client display matches the backend

Given a real ticket for each winning class

When I reveal it in the client and compare to the backend payload
Then the amount, symbols, and free-ticket flag should match exactly

Scenario: A losing ticket shows no win indicators

Given a confirmed non-winning ticket
When I reveal it in the client
Then no win animation, amount, or free-ticket indicator should appear

Scenario: Client-side tampering can't change the outcome

Given a ticket being revealed normally
When I intercept and try to modify the request before the result shows
Then the displayed outcome should stay unchanged